

Complete Bounded Lozi Dynamics: Variable Stability and the Primitive Orbit Atlas

HCS-C385 theorem and reproducibility package

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Abstract

For the area-preserving Lozi family $F_a(x, y) = (1 - a|x| - y, x)$ with $a > 4$, we reconstruct the entire set of bounded two-sided orbits by a uniform contraction on sequence space. Every binary sequence occurs once, and the whole bounded set stays a positive distance from the nonsmooth switching line. This resolves global completeness, not merely a finite periodic prefix. Projective cone contraction gives the invariant stable and unstable lines. Every period, primitive cycle, signed reciprocal multiplier and reversal then follows with an exact affine-return formula.

Keywords: Lozi map; bounded orbit; symbolic coding; uniform hyperbolicity; primitive cycle; instability clock.

中文摘要

本文对参数大于四的保面积 Lozi 映射，利用双向序列空间上的一致压缩，重建全部有界轨道，并证明每个二进制符号序列恰好对应一个状态。有界不变集与不可微切换线具有严格正间隔，因而全局完备性不依赖有限周期外推。进一步从射影锥压缩构造稳定与不稳定方向，得到所有周期、本原轨道、带符号互逆乘子和时间反演的精确分类。

关键词：Lozi映射；有界轨道；符号编码；一致双曲性；本原周期；不稳定性时钟。

1 A global question beyond finite word enumeration

Finite sign words can be solved exactly without showing that every infinite itinerary exists or that every bounded geometric orbit has been found. For a nonsmooth map, a coding must also resolve derivatives at its branch boundary. We close these questions in one explicit sufficient chamber of the globally defined map

$$F_a(x, y) = (1 - a|x| - y, x), \quad a > 4. \quad (1)$$

Its inverse is $(x, y) \mapsto (y, 1 - a|y| - x)$. Let K_a be the set of points whose full two-sided orbit is bounded. The source clock is one application of (1). We do not identify that clock with any later suspension.

Lozi symbolic dynamics and kneading methods have established owners [1, 2]. Our contribution is a directly proved, normalized chamber theorem with independent exact evidence; no priority for the horseshoe mechanism or cone method is claimed. Earlier project C116 certified a pruned finite prefix at $(a, b) = (2, 1/2)$ of $(1 - a|x| + by, x)$. It did not prove the present statement. The constant-diagonal affine horseshoe in C127 also does not provide the word-dependent derivative matrices below.

2 All bounded two-sided orbits

Put $\Sigma = \{-1, +1\}^{\mathbb{Z}}$, with the left shift σ , and define

$$R = \frac{1}{a-2}, \quad q = \frac{2}{a}, \quad \delta = \frac{a-4}{a(a-2)} > 0.$$

For $s \in \Sigma$ let

$$(\mathcal{I}_s x)_j = \frac{s_j}{a}(1 - x_{j-1} - x_{j+1}) \quad \text{on } \ell^\infty(\mathbb{Z}).$$

Theorem 1. *There is a homeomorphism $\pi_a : \Sigma \rightarrow K_a$ satisfying $F_a \pi_a = \pi_a \sigma$. Every coordinate in the coded orbit obeys $\delta \leq s_j x_j \leq R$. These are all bounded two-sided orbits in the plane.*

Proof. The operator $\mathcal{T}_s$ contracts by $q < 1$ and preserves the closed radius- R ball because $(1 + 2R)/a = R$. Its unique fixed sequence satisfies

$$s_j x_j = \frac{1 - x_{j-1} - x_{j+1}}{a} \geq \frac{1 - 2R}{a} = \delta.$$

Hence its signs are exactly s_j and it solves $x_{j+1} = 1 - a|x_j| - x_{j-1}$. Conversely, a bounded orbit with $M = \sup_j |x_j|$ satisfies $aM \leq 1 + 2M$, so $M \leq R$. The same sign bound applies, and contraction uniqueness recovers that orbit from its sign sequence.

Set $\pi_a(s) = (x_0, x_{-1})$. Starting the contraction at zero, the N th approximation to either coordinate depends only on a finite symbol window and has error at most Rq^N . This proves continuity in the product topology. A state determines the full orbit under the continuous inverse of F_a , so distinct sign sequences cannot map to the same state. Exhaustion gives surjectivity. A continuous bijection from compact Σ to its Hausdorff image is a homeomorphism; the recurrence proves shift equivariance. $\square$

The coding is also uniform in parameters: comparing the fixed-point equations for $a, b > 4$ yields

$$\|x_a(s) - x_b(s)\|_\infty \leq \frac{|a - b|}{(a - 2)(b - 2)}. \quad (2)$$

Indeed, after moving the $2/a$ difference term to the left, the remaining bound is $|a^{-1} - b^{-1}|(1 + 2/(b - 2))$. The chamber boundary is deliberately not called sharp: at $a = 4$ this argument has zero sign margin and supplies no conclusion.

3 Variable-matrix hyperbolicity and complete primitive data

Define $r = (a - \sqrt{a^2 - 4})/2$, so $a - r = r^{-1}$ and $0 < r < 1$. On the smooth sign branch s , the derivative is

$$A_s = \begin{pmatrix} -as & -1 \\ 1 & 0 \end{pmatrix}, \quad \det A_s = 1.$$

Theorem 2. *The set K_a is uniformly hyperbolic. The derivative has continuous transverse invariant lines, with stable forward contraction and unstable backward contraction bounded by r^n in the max norm. For every $n \geq 1$, the whole plane has exactly 2^n points fixed by F_a^n .*

Proof. For $|v| \leq r|u|$, the derivative gives $|u'| \geq (a - r)|u| = r^{-1}|u|$ and $|v'| \leq r|u'|$. The slope map $t = v/u \mapsto (-as - t)^{-1}$ preserves $[-r, r]$ and contracts by at most r^2 . Composing from the infinite past gives a uniformly convergent continuous unstable slope. The inverse matrices preserve the cone $|u| \leq r|v|$; composing their slope maps from the infinite future gives the stable line with inverse-time expansion at least r^{-1} . The two cones meet only at zero. In each cone its dominant coordinate equals the max norm, giving the stated estimates with constant one. Theorem 1 keeps K_a away from $x = 0$, so these are actual derivatives, not derivatives assigned to singular points. The conjugacy gives 2^n periodic states. Every periodic orbit is bounded, so none has been missed outside K_a . $\square$

For a based word $w = s_0 \dots s_{n-1}$, compose in forward time:

$$M_w = A_{s_{n-1}} \dots A_{s_0}, \quad c_w = \sum_{j=0}^{n-1} A_{s_{n-1}} \dots A_{s_{j+1}} \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

Empty products are identities. Hyperbolicity makes $I - M_w$ invertible, and the unique based state is $(I - M_w)^{-1} c_w$. Its least period is exactly the least repetition period of w . Therefore the number of oriented primitive cycles is

$$P_n = \frac{1}{n} \sum_{d|n} \mu(d) 2^{n/d}.$$

Let $\tau_w = \text{tr } M_w$. The two real multipliers are reciprocal, with

$$|\lambda_u(w)| \geq r^{-n}, \quad \text{sign } \lambda_u(w) = (-1)^n \prod_j s_j, \quad \lambda_s(w) = \lambda_u(w)^{-1}. \quad (3)$$

The sign follows from the sign of each dominant-coordinate factor $-as_j - t_j$. Repeating w raises M_w and its multipliers to powers. In particular the repeated traces satisfy $\tau_{k+1} = \tau_w \tau_k - \tau_{k-1}$, $\tau_0 = 2$, $\tau_1 = \tau_w$.

The involution $J(x, y) = (y, x)$ obeys $JF_a J = F_a^{-1}$. It reverses primitive words up to rotation and conjugates a return matrix to its inverse. We retain both orientations rather than dividing the primitive count by reversal. Branchwise determinant one and the global piecewise-affine bijection also give area preservation.

4 Conclusion

The contraction theorem exhausts the global bounded set and resolves the switching-line issue in an explicit sufficient chamber. Variable signed stability, primitive repetition and reversal are then controlled for every period, rather than guessed from a prefix.

References

- [1] M. Misiurewicz and S. Štimac, *Symbolic dynamics for Lozi maps*, arXiv:1601.01745 (2016). Primary preprint.
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Round one: hyperbolicity and primitive atlas.